

Concurrency & Distributed Systems

Week 1 — Part 4: Measurement-Driven Engineering:
EnhancedLoadClient

Dr. Mohamad Aoude

Lebanese University
Faculty of Engineering

May 11, 2026

Demo is Not Enough

Previously:

- We ran a simple LoadClient
- We observed “it feels slower”
- We guessed performance

Problem

Guessing is not engineering.

Goal

We now measure:

- p50 latency
- p95 latency
- Throughput (requests/sec)

Latency Percentiles

p50 (Median)

- 50% of users are faster than this
- Represents “typical experience”

p95 (Tail)

- 95% of users are faster
- Slowest 5% determine perceived quality

Critical Insight

High throughput with bad p95 = bad system.

Throughput

$$\text{Throughput} = \frac{\text{Total Requests}}{\text{Total Time}}$$

If:

- 500 requests completed
- in 2 seconds

$$\text{Throughput} = 250 \text{ req/sec}$$

Remember

Latency measures individual experience. Throughput measures system capacity.

Step 1: Start Server

Terminal 1:

```
javac ImprovedMultiThreadedServer.java  
java ImprovedMultiThreadedServer
```

Step 2: Compile Client

Terminal 2:

```
javac EnhancedLoadClient.java
```

Step 3: Run Experiments

Light load:

```
java EnhancedLoadClient --clients=10 --  
requests=10
```

Medium load:

```
java EnhancedLoadClient --clients=50 --  
requests=20
```

Heavy load:

```
java EnhancedLoadClient --clients=200 --  
requests=20
```

Single-Threaded Server

Expected behavior:

- Latency grows linearly
- p95 extremely high
- Throughput capped
- CPU mostly idle

Q

Queue formation dominates.

Multi-Threaded Server

Expected behavior:

- Latency collapses initially
- Throughput increases significantly
- CPU rises moderately

Then push further:

- p95 increases again
- Throughput plateaus
- CPU spikes

Why Too Many Threads Hurt

Three causes:

- Context switching overhead
- Memory overhead (stack per thread)
- Lock contention

$$\text{Optimal Threads} \approx \text{cores} \times \left(1 + \frac{\textit{Wait}}{\textit{Compute}} \right)$$

Lab 1.1 Requirements

Students must submit:

- Measurement table (p50, p95, throughput)
- At least 3 pool sizes tested
- CSV export file
- 5–8 sentence engineering explanation

Required Measurement Table

Load	p50	p95	Throughput
Light (10)			
Medium (50)			
Heavy (100)			

N

o numbers = no credit.

Students Must Answer

- 1 Why did latency drop with threads?
- 2 Why did p95 increase under heavy load?
- 3 Why didn't throughput grow infinitely?
- 4 Was the system I/O-bound or CPU-bound?

Engineering Language Required

Use:

- queue
- overlap
- scheduling overhead
- saturation

What This Demonstrates

- Concurrency hides waiting.
- Concurrency does NOT create CPU power.
- Tail latency reveals overload.
- Measurement drives architecture decisions.

I

f you cannot measure it, you cannot engineer it.

What happens when threads share mutable state?

N

ext Lecture: Race Conditions, Visibility, and Deadlocks.